Prompt Engineering for Data Augmentation and Machine Translation Evaluation in Modern Greek-English

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1 Introduction

This paper focuses on dataset creation and evaluation metrics in the scope of machine translation, with Modern Greek to English as the language pair. Given the unique linguistic challenges present in Modern Greek, including flexible syntactic structures, idiomatic expressions, and diminutives, machine translation for this language pair is challenging. To address these challenges, we used the Tatoeba challenge dataset as a foundation and augmented it through data augmentation techniques that involve large language models and ChatGPT. We then applied and compared a range of evaluation metrics, including BLEU, COMET, and GEMBA, to assess the quality of the machine translation models.

2 Related Work

This paper was inspired by the *Machine Translation* chapter of *Speech and Language Processing* (Jurafsky and Martin, 2018), which refers to both statistical and neural approaches, as well as translation models, evaluation metrics, and the shift from phrase-based methods to neural machine translation. In *Data Augmentation using LLMs: Data Perspectives, Learning Paradigms and Challenges* (Ding et al., 2023), various studies are examined concerning data augmentation, including an exploration of learning paradigms where the augmented data are used for further training. In *Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science* (Bender and Friedman, 2021), long form data statements are proposed as a professional practice for dataset creation and publication, as they address scientific and ethical issues that arise when the dataset origin and purpose are unknown.

In *Do Prompt-Based Models Really Understand the Meaning of Their Prompts?* (Webson and Pavlick, 2022) the effectiveness of prompt-based learning in zero-shot and few-shot scenarios is investigated, challenging the assumption that task instructions are understood in a human-like manner, as models can perform well even with irrelevant or misleading prompts. In *Towards Making the Most of ChatGPT for Machine Translation* (Peng et al., 2024), the optimization of ChatGPT's performance in machine translation tasks is explored, by adjusting temperature settings and introducing task specific and domain specific prompts; time and resources allowing, they could be proven useful in further analysis of prompting techniques.

3 Dataset Description

This dataset, developed by Martha Koukourikou and Melina Paxinou, is based on the Tatoeba challenge (Tiedemann, 2020) and was augmented for the purposes of the experiments in this paper. It includes conversational texts with examples of direct speech, casual expressions, and simple sentence structures, covering a variety of everyday topics. The dataset reflects key linguistic phenomena such as flexible syntax, idiomatic expressions, and culturally specific vocabulary, making it particularly relevant for machine translation tasks involving Modern Greek spoken in casual settings.

4 Linguistic Analysis

4.1 Challenging Linguistic Phenomena in Modern Greek

4.1.1 Emphatic syntactic structures

Greek syntax follows the SVO pattern, however, it can be very flexible. In Greek, it is very common to omit the subject, as the verb form indicates the first, second, or third person, as well as singular or plural form. For example, the sentence "Εγώ διαβάζω ένα βιβλίο." can also be written as "Διαβάζω ένα βιβλίο."

“Εγώ διαβάζω ένα βιβλίο.”
1SG.NOM read.1SG INDEF book.ACC
SUBJ V OBJ
“I am reading a book.”

“Διαβάζω ένα βιβλίο.”
read.1SG INDEF book.ACC
V OBJ
“(I) am reading a book.”

While the basic syntactic structure in Greek follows this pattern, it is often changed to emphasize a different part of the sentence. Specifically, we can say “Ένα βιβλίο διαβάζω.” and place the object at the beginning of the sentence to emphasize that it is a book that I'm reading.

“Ένα βιβλίο διαβάζω.”
INDEF book.ACC read.1SG
OBJ V
“It is a book that I am reading.”

This phenomenon can be challenging in translation, because this versatile change in position can contribute to the actual meaning of the sentence. English does not have the same versatility and, in order to convey the same meaning, a human translator would have to add extra words (e.g. “It is a book that I'm reading.”) or omit the emphatic structure entirely if it does not play an important part in this specific context (e.g. “I'm reading a book.”).

In testing the model, the model could be given sentence pairs where the only difference is the word order, such as the sentences mentioned above. This could give us some insight into how the model understands the different syntax forms and what meaning they convey. For example, by giving it sentences where the subject is omitted, we can see if it can identify the person and number, and by giving it sentences where the object is fronted, we could see if it understands the different nuances in different versions of the same sentence.

4.1.2 Diminutives

Diminutives are quite commonly used among Modern Greek speakers. They usually signify that something is smaller than the original, e.g. “τραπεζάκι.” Here, the original word is “τραπέζι” and the suffix is “-άκι,” which literally means “small table.”

“τραπεζ-άκι”
table.DIM
“small table”

However, diminutives often give additional meaning to words which is not necessarily “small.” For example, “αντράκι” comes from the Greek word for man (“άντρας”) and refers to a man who is brave. Although it has the diminutive “-άκι,” it does not refer to a little man and has positive connotations. Similarly, “γυναικάκι” comes from the Greek word for woman (“γυναίκα”) and refers to an insignificant woman that does not deserve attention. It has negative connotations and does not mean “little woman.”

“αντρ-άκι”
man.DIM
“brave young man”

“γυναικ-άκι”
woman.DIM
“insignificant woman”

A human translator would be able to understand the difference between the nuances of the diminutive “-άκι.” They would understand that the first case would be translated as “small table,” as there is no ambiguity in its meaning. Depending on the context, they could even omit the translation of the diminutive if the reader can understand that the object is small (e.g. “coffee table” for “τραπεζάκι σαλονιού” instead of “small table for the living room”).

“τραπεζ-άκι σαλονιού”
table-DIM living.room.GEN
“small table for the living room”

For “αντράκι,” they could choose a descriptive approach depending on the context, such as “brave young man,” as there is no exact equivalent in English. As for “γυναικάκι,” they could either use the word “cipher” or turn to a more descriptive approach, such as “insignificant/worthless woman,” as in the first case the meaning of “woman” would be lost.

A machine translation model would probably understand the meaning of “small” in this example and would reproduce “small table” as a potential translation. However, it would fail to understand the nuances of "αντράκι" and "γυναικάκι," as the positive and negative connotations respectively are expressed with the use of the same diminutive. Therefore, providing the machine translation model with more examples on the different uses of such diminutives, as well as combining words with diminutives with the surrounding words could help us see if the model generates translations that better capture these subtle differences.

4.1.3 Idioms

Modern Greek is one of the richest languages in the world and includes several idioms that are used by native speakers on a daily basis. Idioms are especially hard to translate, as the translator either has to find an equivalent in the target language or translate it literally and lose the particular wording of the phrase. An example of a common idiom would be "Βρέχει καρεκλοπόδαρα.", which has the equivalent English translation "It's raining cats and dogs." This would not be particularly difficult for a translator, as the two phrases are equivalent and have no difference in meaning whatsoever. On the contrary, a machine translation tool would most likely not understand that this is an idiom, and it would translate it word-for-word.

"Ρίχνει καρεκλοπόδαρα."
pour.3SG chair-leg.PL
V OBJ
"It is pouring chair legs."

Nonetheless, Modern Greek has several idioms that do not necessarily have equivalents in English. For example, "Έχει πολύ ψωμί η υπόθεση." means that there is a lot going on in a particular situation. This would pose a challenge to both a translator and a machine translation tool, as both would need to translate it without the use of an equivalent idiom. Specifically, a human translator could use the phrase "This situation is very juicy." if the context allows, or they could simply translate it as "This situation is very interesting."

"Έχει πολύ ψωμί η υπόθεση."
have.3SG much bread the.F.SG.NOM case
V OBJ SUBJ
"The case has a lot of bread."

Machine translation tools often ignore the use of idioms and translate such sentences literally, in this case as "The case has a lot of bread." The model's performance could be evaluated by prompting it to translate several such idioms and compare it with the same idioms translated by humans. The model could also work with context, as the surrounding words should signify that we are not actually talking about chair legs or bread.

4.1.4 Words without equivalents

Modern Greek has several words that have no equivalents in English. Some of those words are:

"έρωτας"
love
romantic, sexual love

"φιλότιμο"
friend-honor
pride, honor, sense of duty

"μανία"
frenzy
frenzy, compulsion, obsession

These words pose a challenge in translation, as they would have to be described with more than one word in English. The meaning that has to be conveyed for each of these words respectively is "romantic and sexual love that surpasses simple physical attraction", "pride, honor, sense of duty", and "fury, frenzy, compulsion, obsession, manic state".

When human translators tackle this kind of words, they either describe them or choose the closest equivalent, in this case "love", "pride", and "obsession" or "manic state" respectively, depending on the context. However, such words are not only difficult to translate but can also convey cultural meanings that are lost after the translation process. Words such as "φιλότιμο" are deeply rooted in the speaker's culture and shape the Greek identity, thus making it crucial to consider both linguistic and cultural contexts when translating it.

A solution for evaluating the translation of this kind of words by a machine translation model would be to check if it handles the words similarly to a human translator, thus providing the closest possible meaning that exists in the target language. Similarly to a human translator, in cases where an absolute equivalent does not exist, the model should be able to provide us with the general idea of what the source text's meaning is, even if it is not the most accurate translation. However, the issue of accuracy would arise, as sometimes the distinction between subtle differences in word meaning (e.g. romantic/sexual love and other types of love) could be crucial for the outcome of the translation.

Ultimately, preserving the unique nuances of such words requires a balance between conciseness and cultural sensitivity, which poses an ongoing challenge for both machine translation models and human translators.

4.1.5 Gender

Modern Greek is a highly gendered language. Gender influences many aspects of the language, such as verb conjugation, noun forms, adjective and article agreement, and is context dependent. In practice, this means that in sentences where the subject is omitted, e.g. "Διαβάζει ένα βιβλίο.", we do not actually know if the subject is masculine or feminine without having any context.

"Διαβάζει ένα βιβλίο."
read.3SG INDEF book.NEUT.SG.ACC
V OBJ
"(Someone) is reading a book."

When handled by a human translator, this phrase would take the subject of a previous sentence based on context, e.g. "Martha bought a book. She is reading a book." However, when looking at this sentence as a standalone, there is no way of knowing if the subject is masculine or feminine. A human translator would have many ways of approaching this. Firstly, a translator could use "they" as a pronoun, thus making the sentence gender neutral, even though this would compromise the meaning of the singular form of the verb. Another way they could handle this is by using "he", which is the most commonly used pronoun when faced with ambiguity. Lastly, they could find a descriptive way of translating this sentence, such as "Someone is reading a book." or "This person is reading a book." This approach maintains a gender-neutral form but adds words that did not exist in the original sentence.

As for noun forms, a common example would be "φίλος" and "φίλη". Even though both words mean "friend", there is no distinction between male and female friend in English. This can also cause issues during translation, as the same text could contain both words and the difference between them could be important to the overall meaning.

"φίλος"
friend.M
friend

"φίλη"
friend.F
friend

A human translator would most likely handle this simply using the word "friend" in both cases. Most of the time, the reader can understand the gender of a person by the context, so it would not be an issue to translate both as "friend".

When it comes to machine translation, models would most likely translate both as "friend". Especially in sentences where gender does not matter, it is a relatively easy task. As mentioned above, though, it would be proven difficult for a model to translate a larger text that refers to both male and female friends, and this distinction is crucial for portraying the meaning of the text. Again, a more contextual approach when evaluating a machine translation model would be important, as it would help determine if the model understands the referents in each case.

When faced with no context, then the model should maintain the singular form and not use "they". This is not an ideal approach, as the gender of the subject is unknown, but adding words (e.g. "someone" or "a person") is not a good solution either. Therefore, choosing "he" or "she" as a default pronoun would be the most accurate translation.

4.2 Dataset

4.2.1 Diminutive suffixes as a morphological property

Understanding whether certain suffixes express specific meanings or sentiments or hold an arbitrary meaning altogether could be proven helpful as a property, as diminutives not only indicate that an object is "smaller," but also carry the potential to alter the overall sentiment of a sentence positively or negatively.

This property was inspected with the use of spaCy, an open-source library for natural language processing (NLP). However, spaCy's functions are not as effective for Modern Greek, so we limited its use to tokenization. After tokenizing both the source and target sentences, and separating punctuation, we manually compiled a list of potential diminutive suffixes in Modern Greek, along with their possible inflections. We used a for-loop to iterate over every token in the dataset, and, if it contained one of our potential endings, it was added to a list and was printed along with the source and target sentence in which it appeared. We also added a condition to the loop to ensure that the token had more characters than the suffix, as sometimes the suffix itself could appear as a separate token.

Our findings were as follows:

i. Some diminutives had direct equivalents in English, such as *αυτοκινητάκι* ("toy car").
ii. Some words, although originating from a combination of a base word and a diminutive ending, are established as single words in Greek (e.g., *κοτόπουλο* for "chicken").
iii. In some cases, the diminutive meaning was lost in translation (e.g., *καφεδάκι* translated simply as "coffee").
iv. Some diminutives were translated using more than one word, such as *κοριτσάκι* ("baby girl").
v. In certain instances, the diminutive carried an additional nuance or connotation, such as *δασκαλίτσα*, which translates to "teacher" but carries a derogatory meaning.

The MT model could learn the possible suffixes of diminutives and the difference between words with the same character length and the suffixes themselves. As there is no open-source library that identifies diminutives for Greek, a manual annotation process would clarify if the tokens in the list we compiled consist of diminutive suffixes. Potential features for a machine translation model could include the presence of a diminutive and whether it carries a positive or negative connotation, represented as true or false statements.

To augment the dataset with sentences that contain diminutives, a prompt must be given to an LLM, which should contain the desired sentence length (in tokens), the condition that the sentence should contain a diminutive, the list of possible diminutive suffixes, whether a diminutive should hold a positive, negative, or neutral meaning, and the desired format with an example.

4.2.2 Lack of subject as a sentence-level property

The significance of this property in Modern Greek is immense, as a quite common feature of the language is the omission of the subject when it is deemed unnecessary.

After testing the POS (part-of-speech) and syntactic parsing functions of Stanza and spaCy on our Greek dataset, we found the results to be unreliable. As a result, we used a Greek NLP toolkit that, after manual inspection, seemed to be more accurate. Thus, we identified the subjects and verbs in both Greek and English (using the Greek NLP toolkit and spaCy respectively) (NLP-AUEB, 2024).

Taking into account that some of the POS and syntactic tags might be false, as they were tagged using the aforementioned libraries, we calculated that the subject-to-verb ratio of the Greek sentences is 0.35 and the subject-to-verb ratio of the English sentences is 0.79. This confirmed our suspicions that it is quite common for Greek sentences to lack a subject, which can negatively impact the precision of machine translation. This can also be seen in Figures 1 and 2, which show the sentence length distribution across the dataset. The Greek sentences in our dataset have a slightly shorter length in tokens, due to the pro-drop phenomenon.

The omission of the subject has a direct impact on machine translation. Not knowing the subject can lead to misidentifying the gender of the subject or using the wrong verb aspect. It can also lead to biased or completely false results, error propagation, and coherence issues.
Potential features could involve the lack of subject, the correlation of verb suffix with tense and pronoun, and, where context allows, retrieving the subject based on coreference. However, the issue of gender identification in the third person would remain present.

Possible dataset augmentation prompting could include features, such as: omission of subject in source language, ambiguity in gender and/or aspect, English translations containing all possible persons/aspects. The desired sentence length (in tokens) and the desired format with an example should also be included.

4.2.3 Semantic compression in words without direct equivalents as a property above word level

Using semantic expansion or compression can assist in translating languages like Modern Greek, which contains many words without direct English equivalents.

To explore this, we identified stems of difficult-to-translate words and iterated through the dataset. While inspecting our dataset we observed that our list with words without direct equivalents does not only contain words that do not exist as a concept in English, but also concatenated words that need to be described when translated. We printed each of the words with the corresponding source and target sentences and observed their translations.

Our findings included the number of untranslatable words in our dataset (70, containing repetitions) and the amount of words used to translate them. The source-to-target word ratio is 1:1.48, which leads us to the observation that it is quite common to describe the meaning of more complicated words, in order to get their meaning across. However, we did notice that single-word translations often lose nuance (e.g. *μεράκι* as "passion"), compared to described translations that almost entirely retained their meaning (e.g. *καφενείο* as "traditional café"). Culturally specific words, such as *τσίπουρο* (i.e. a Greek alcoholic drink), were often simply transliterated. Therefore, we can see that semantic compression is prevalent in the translations of our dataset. Similarly to the previous property, this is supported by Figures 1 and 2, which demonstrate the slightly longer average word count in English sentences.

An MT model might only capture part of the meaning of such words, rather than the full concept. For example, when dealing with concatenated words (e.g. *ξανα-*), the model could learn the correlation between the affix *ξανά* translating to "again," based on context. However, in the case of conceptually non-existent words, the model might not find a one-to-one correspondence, leading to the challenge of indeterminacy in translation. The lack of exact translations between languages, where some concepts exist in one language but not in another, requires the model to learn the closest possible translation based on context, rather than use an exact equivalent.

To prompt an LLM for dataset augmentation purposes, we could include features such as: a list of possible words without direct translation equivalents (e.g. *φιλότιμο*, *μεράκι*), desired sentence length (in tokens), and desired format with an example. In this case, the translation strategy should also be specified, depending on whether we are aiming for conceptually unique words that are usually translated with simpler equivalents or words that can be described with longer phrases instead of single-word translations.

5 Prompting with LLMs

Table 1: Dataset Before and After Augmentation

Statistic	Before	After
Dataset (sents)	954	832
Sentence length	El: 6.78, En: 7.53	El: 13.68, En: 14.74
Diminutives	19	138
WO[1]	1	42
S/V ratio[2]	El: 0.36, En: 1	El: 0.35, En: 0.79
WWE[3]	17	70
Idioms	8	33

5.1 Data augmentation

To create a balanced dataset that includes all our phenomena in their different forms, we worked on data augmentation with LLM prompting. At first, we identified which of our phenomena were underrepresented and used ChatGPT to create artificial data. Our augmentation was not at all quantitative, but qualitative. As the original dataset from the Tatoeba challenge only contained short and simple sentences, we gave the existing sentences to ChatGPT and instructed it to generate longer text with richer vocabulary and correct use of the Greek language, using one of the phenomena we were analyzing. In this way, we ensured that we had various forms of diminutives, idioms, and words without direct translation equivalents, including their different meanings and translations, as well as sentences that included some of the same words but without the diminutives or idioms for comparison. Lastly, we did not have to do anything for sentences with lack of subject, as we already had plenty in our dataset.

[1] WO: Changes in Word Order.
[2] S/V ratio: Subject-to-verb ratio metric.
[3] WWE: Words Without Equivalents.
[4] All prompts mentioned are available on VUgit.

We started with simple prompts, which followed a zero-shot prompting technique.[4] They only contained instructions on how to create new sentences based on existing ones with a greater word count and an example for reference. Then, we added the tab-separated form of our dataset, so that ChatGPT gives us the desired output directly. Finally, we refined the prompt so that it also contains the number of sentences, register, vocabulary, and integration of one of our phenomena. This final prompt form was a combination of prompt chaining and few-shot prompting (Prompt Engineering Guide, n.d.), as it contained detailed instructions and gave ChatGPT a specific task, while providing numerous examples (see Section 8 in the Appendix for the final prompts). Few-shot prompting is a more beneficial technique compared to zero-shot prompting in this case, as the latter only relies on the clarity of our instructions and the existing data it already has from pre-training, whereas few-shot prompting contributes to the model's understanding and pattern recognition (e.g. how diminutives are used in sentences). All our prompts contained lists of already existing sentences that we needed to lengthen, except one prompt, which contained a list of Greek idioms and a request to create sentences from scratch based on the idioms of that list. For the LLM to give us sentences with different word orders but the same meaning, we had to manually create a few instances as examples, as the model itself could not understand how the difference in word order slightly changed the nuances of our sentences.

As Modern Greek is a medium resource language, we were able to perform a fairly good data augmentation with ChatGPT. We evaluated translations based on the following criteria: grammar, syntax, whether or not the phenomenon we were investingating was captured correctly, fluency, and compliance with prompt guidelines (e.g. sentence length).

Our observations were promising; however, we noticed a few areas where ChatGPT generated sentences and their respective translations were lacking. Although grammar and syntax were mostly correct, the model sometimes did not implement our phenomena correctly. For instance, it added diminutives in words that do not take diminutives or added the wrong type of diminutives to them. Creating sentences with the same meaning but different word orders depending on the emphasis was its greatest challenge, as it was where we got the most errors and where we needed a substantial number of examples to instruct it sufficiently.

Our new dataset is represented in Table 1, together with our previous dataset. The average sentence length has doubled in size, and the instances that contained our phenomena have increased substantially. Sentences with diminutives have risen from 19 to 138, and idioms from 8 to 33. It is important to note that when we calculated the subject-to-verb ratio for the Greek and English sentences in our datasets, we noticed that the difference was less prominent. That can easily be explained by the fact that the new dataset is substantially more complex, while the previous one mostly followed the SVO pattern with little variation.

5.2 MT Evaluation

For our machine translation output evaluation, we used the GEMBA prompt template to create our ChatGPT prompts. GEMBA is a ChatGPT-based metric that aims to evaluate translation quality. Initially, we simply used the template as is to test it, giving it the source, reference, and translated segments and a scoring scale of 0 to 100. Then, we modified it slightly to give us more sentences at the same time by adding three lists of sentences: Greek sentences, English translation references, and English translations (see Section 8 in the Appendix for the final prompt). This method was quite useful in terms of speed, as we were able to analyze about 30 sentences per request. For our final refinement, we added certain criteria for evaluation, such as grammar, syntax, fluency, and whether our specific phenomenon is captured correctly. In that way, we managed to evaluate translations for each of our phenomena while getting short justification points for each result.

To ensure that our results were valid, we added an extra verification step; using a second account of a completely different ChatGPT user. As shown in Table 2, for the same prompts, sentences, criteria, and phenomena, two different users received two completely different results. User 1 constantly received high results, while User 2 significantly lower. Each user asked ChatGPT to justify its answers, or even conversational prompts, such as "Are you sure?", and the LLM gave fairly accurate explanations where the MT output was incorrect in each sentence. However, the difference in scores did not change.

6 Machine Translation Task

6.1 Experiments

For the Machine Translation task, we used CTranslate (OpenNMT, n.d.), a python and C++ library that facilitates interaction with Transformers, and M2M100 (Hugging Face, n.d.), a multilingual encoder-decoder model that supports Modern Greek. Our goal was to translate challenging sentences from our dataset and evaluate the output with metrics such as BLEU (Papineni et al., 2002) and COMET (HuggingFace, n.d.), as well as to prompt an LLM with the GEMBA prompting template (Wang, 2024) to provide an evaluation for us. We also calculated the cosine similarity between source-reference and source-target sentences to compare. We aim to evaluate how the model handles various linguistic phenomena identified in our analysis and to examine the interaction between different evaluation metrics and the machine translation output.

6.2 Results

The data originating from the augmentation with ChatGPT provided a diverse and enriched dataset containing linguistic phenomena that characterize the Greek language. This allowed for a targeted evaluation using all these different metrics, highlighting the model's strengths and weaknesses in handling complex translation challenges.

BLEU measures the overlap between translated segments and reference segments using n-grams. Although it can capture some similarity between the two segments, it does not take semantics into account. As seen in Table 2, idioms and words without equivalents have the lowest BLEU scores, which reflects the quality of our translations (e.g. *Παρά τρίχα*, an idiom that means "That was close", is translated as "Despite hair"). However, we did notice several extremely low scores; the sentence *Το νοίκι πρέπει να πληρώσω.* was translated as "I have to pay the rent." and the reference translation was "It's the rent that I need to pay." Although there is some nuance difference between the two sentences, the MT output is fairly good. However, it got an extremely low BLEU score (7.1e-155), as the n-grams of the two segments do not overlap. Compared to the current state-of-the-art (SOTA) in Modern Greek-to-English machine translation (Papadopoulos et al., 2024) with a BLEU score of 79.3 (on a scale of 0 to 100), the BLEU scores obtained in our project were significantly lower. Notably, this SOTA score is based on the machine translation performance of the original dataset on which our dataset was based prior to augmentation, making it a valuable benchmark for comparison in our study. Therefore, the BLEU score can be useful in simple sentences with basic syntactic structure, as these can be translated literally and follow similar n-gram patterns; however, it is not useful for more complex translation evaluation.

Next, we used Comet for comparison to BLEU. We used *wmt20-comet-da*, an evaluation model that uses word embeddings to calculate similarity. Considering that it has a wider score range than BLEU, its evaluation results were higher. Although we had some good translations, high scores do not reflect the majority of our data. However, we can see that the two evaluation systems agree on the fact that idioms and words without equivalents had the lowest translation quality, followed by diminutives.

As an additional evaluation metric, we created sentence embeddings with *bert-base-multilingual-cased* to find the cosine similarity of source-translation and source-reference sentence pairs. This method helps to assess the semantic closeness of the machine translation to the reference. By comparing these similarities, we could evaluate the quality of the translation by checking how well the translated English captures the meaning of the source Greek relative to the reference translation. In this case, the results were surprising. Our findings supported the notion that the similarity of source-translation sentences was higher than that of source-reference. This can be explained by the fact that the mean sentence vectors in MT are computed based on all token vectors individually, and since MT often produces more literal translations while preserving the source syntax, the similarity tends to be higher. However, the differences were very small and all similarities were approximately 0.5.

Lastly, even though our ChatGPT evaluation results were inconsistent, the LLM had useful insight into why each translation is incorrect. It could understand the errors in each case and explain them with very high accuracy. For example, in our sentences with word order other than SVO, ChatGPT could understand where the emphasis was with the use of the English reference, and it even had a strong understanding of Greek idioms. Although its scores were inaccurate, it was proven to be the most useful machine translation evaluation tool we used.

It should be noted that, when ChatGPT was asked the same prompt but without the grading criteria, that is, grammar, syntax, fluency, and faithfulness to the linguistic phenomenon specified, its scores remained inconsistent but tended to be higher than when it was asked the complete prompt. This time, it did not provide us with explanations on why the sentences contain mistakes, however, it gave the lowest scores to the translations of words without equivalents and idioms, similarly to when it was asked the full prompt.

7 Discussion

After obtaining the results of all our evaluation methods, we performed a detailed analysis of our translations to identify and understand the errors. We translated and analyzed a subset of our dataset that we believed accurately represents the key phenomena and provides a comprehensive view of our data.

By examining some of the output of the MT model, we identified several serious errors that, in most cases, make the translations unusable. These errors mainly affected syntax, omissions, and task-specific errors. More specifically, we noticed that the diminutives were completely overlooked by the MT model, which is a great example of both omission and task-specific error. The diminutives were either completely skipped and the stem of the word was translated normally, or they were made into entirely new words (e.g. *γιαγιούλα* for grandma was translated as "youla"). In a similar note, words without direct translation equivalents were either assigned a word with similar pronunciation (e.g. "raki", a Greek alcoholic beverage, was translated as "rack"), or they were completely omitted, along with the part of the sentence in which they were in. Moreover, idioms were translated literally, which consists of a task-specific error (e.g. "I made the sea at work" for "I messed everything up at work"); therefore, the translations in some cases were difficult to understand. In sentences in word order other than SVO, each of the words was literally translated and the model created sentences that did not make any sense (e.g. "The shoes asked them to get out." for "It is their shoes that she asked them to take off."). Although they are syntactical errors, they were not simply syntactically incorrect; in fact, the translations were so bad that a native speaker might not have understood the general meaning of the sentences.

Lastly, the phenomenon with the least amount of errors was pro-drop. The sentences were sometimes assigned a subject based on pronouns that were present later in the text; therefore, many of the translations were correct. However, a serious error was that, in some cases, the subject was made neutral. Despite this, the model was still able to handle pro-drop relatively well compared to other phenomena, though occasional errors in subject assignment indicate room for improvement.

8 Conclusion

Dataset augmentation with large language models shows promise, especially when utilizing few-shot prompting techniques. However, the machine translation output in our experiments was suboptimal, with many translations being non-sensical and unusable. The evaluation scores were both low and inconsistent, revealing the inherent challenges in translating complex linguistic phenomena such as idiomatic expressions and diminutives. These findings emphasize the difficulty of producing high-quality translations for intricate language structures, particularly in language pairs with few shared characteristics, such as Modern Greek and English.

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Table 2: Evaluation Results: mean scores per linguistic phenomenon

Metric	WO	WWE	Idioms	Pro-drop	Diminutives	Range
BLEU	0.68	0.36	0.36	0.50	0.47	0 to 1
COMET	0.59	0.20	0.09	0.52	0.35	-1 to 1
Sentence Vectors Mean Cosine Similarity	REF: 0.48 TRA: 0.50	REF: 0.56 TRA: 0.55	REF: 0.56 TRA: 0.57	REF: 0.59 TRA: 0.59	REF: 0.55 TRA: 0.55	-1 to 1
ChatGPT GEMBA User 1	65.21	45.83	41.25	70.31	71.79	0 to 100
ChatGPT GEMBA User 2	70.83	47.08	47.92	65.00	49.64	0 to 100
ChatGPT GEMBA User 2 without evaluation criteria	70.33	58.33	65.83	71.88	70.36	0 to 100

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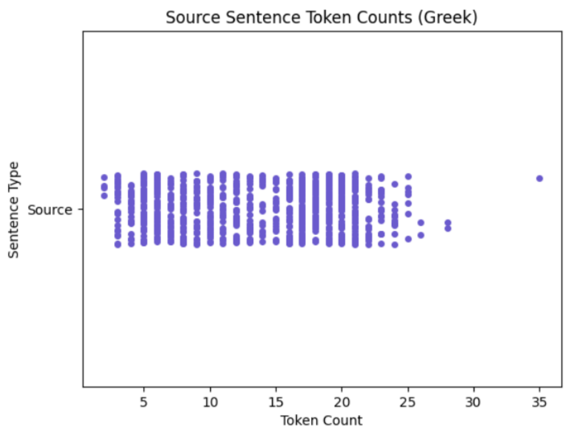

Figure 1: Sentence length in tokens - Greek

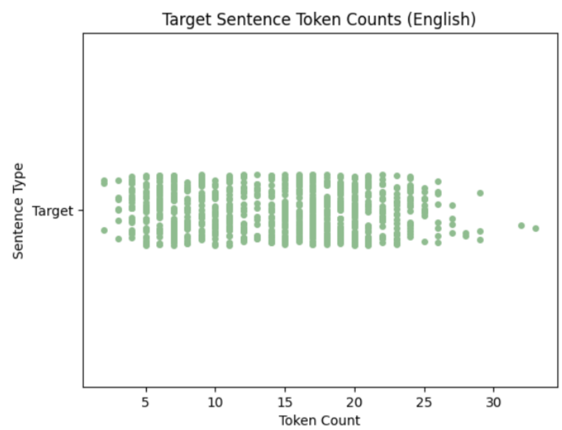

Figure 2: Sentence length in tokens - English

Appendix

Prompts

Data augmentation

Take the following sentences in {source language} along with their {target language} translation, and I want you to do the following: I want each {source language} sentence to consist of at least {number} words and adjust the {target language} translation accordingly. I want you to give me at least {number} sentences. I want the style to be simple, the vocabulary to be varied, and I want half of the sentences to contain {source language linguistic phenomenon} like {source language phenomenon example}. Give me the output in the format specified below and separate them with tab:
{source language code} {target language code}
{source sentence} {target sentence}
The sentences I want you to change are the following:
{List of source sentences}
{List of target sentences}

This is a list of idioms in {source language}:
{List}
Create a sentence for each idiom from this list. Sentences must contain at least {number} words each, one of the idioms above and the language should be simple. Give me the output in the format specified below and separate them with tab:
{source language code} {target language code}
{source sentence} {target sentence}

MT evaluation

Here are three lists, each corresponding to {source language} sentences, their {target language} human references and their target language translations respectively.
{source language sentence list}
{target language reference list}
{target language sentence list}
Score the translations from {source language} to {target language} with respect to the human reference on a continuous scale from 0 to 100, where a score of zero means "no meaning preserved" and score of one hundred means "perfect meaning and grammar".
Example output:
{source language} source: "{source sentence}"
{target language} human reference: "{target reference sentence}"
{target language} translation: "{target sentence}"
Score:
Base your scoring on the following criteria:
1. Grammar
2. Syntax
3. Capturing and incorporating the linguistic phenomenon in the source language sentence (if it is preserved in target language translation in relation to the target language human reference)
4. Fluency of the sentence (if a native speaker would use this sentence

Long Form Data Statement

Dataset Title:
Simple Modern Greek to English Dataset for Machine Translation
Link to dataset: https://git.vu.nl/auf581/lad-mt-el-en

Dataset Curators:
The source dataset was curated by Jörg Tiedemann and collaborators as part of the OPUS project. The specific dataset used was curated by Martha Koukourikou and Melina Paxinou.

Dataset Version:
Source dataset: v2023-09-26
Current dataset: v2024-11-26

Dataset Citation:
Source dataset: Tiedemann, J. (2020). The Tatoeba Translation Challenge – Realistic data sets for low-resource and multilingual MT. Proceedings of the Fifth Conference on Machine Translation, 1174–1182. Association for Computational Linguistics. https://www.aclweb.org/anthology/2020.wmt-1.139

Executive Summary:
The source dataset contains Greek-English pairs and is part of the Tatoeba Translation Challenge, while the current dataset is an augmentation of the most linguistically interesting parts of the source dataset, which also covers phenomena, such as diminutives and idioms.

Curation Rationale:
The original Greek dataset was selected to represent linguistic diversity for machine translation tasks, making it one of the highest resource language pairs in the Tatoeba dataset. For our subset, we curated 832 sentences. Our primary goals were to include relatively more complex sentences, a va-

riety of structures, and sentences that feature the linguistic phenomena we are researching, while simultaneously reducing repetitive patterns of the source dataset. We also made some corrections for small orthographic or syntactic errors.

To augment the original sentences, we used several prompting techniques and worked with ChatGPT. Specifically, we expanded many existing sentences by increasing their length, while preserving the phenomena we are investigating, such as lack of subject. We also generated new sentences similar to the existing ones, emphasizing the phenomena of interest, such as idioms. Additional sentences were created manually to reflect word order issues, which could not be effectively reproduced by the language models.

Language Variety:

The original Greek subset focuses on Modern Greek, which is spoken by approximately 13 million people primarily in Greece and Cyprus, and in smaller Greek communities worldwide. It includes texts in standard orthography and reflects common linguistic phenomena in Greek, such as diminutives, idiomatic expressions, words without direct translation equivalents, lack of subject, and flexible syntactic patterns.

Speaker Demographics:

Source texts were contributed by volunteers on Tatoeba.org, including both professionals and volunteers with expertise in Greek and English. As curators of this dataset, we—a team of native Greek speakers with Master's-level education in Linguistics—collaborated with ChatGPT to augment the data, thus increasing its complexity and real-world applicability.

Annotator Demographics:

Martha Koukourikou and Melina Paxinou, Greek post-graduate students with an educational background in Linguistics and Translation.

Text Characteristics:

The dataset consists of conversational texts, with several examples of direct speech, casual expressions, and simple sentence structures. The sentences cover a range of everyday life topics and include all the previously mentioned phenomena that are investigated. Its vocabulary is simple yet contains some culturally specific words that could be proven useful during the machine translation process.

Data Format:

The original data are in.txt format. The current subset is provided in .tsv format, due to the commas in the running text, which excluded the possibility of a comma-separated .csv format.

Additional files

The code used for CTranslate, M2M-100, dataset analysis, BLEU and Comet scores, and cosine similarity is available on VUgit.